

Privacy-Preserving Synthetic Student Attendance Analysis and Prediction System

Final Project Report

Department	Computer Engineering & MCA
Year / Semester	Third Year — Fifth Semester (CE) · Final Year — Third Semester (MCA)
Dataset	Fully Synthetic — 205 Students, 4,100 Records
Report Date	August 2026
Project Status	Phases 0–5 Complete

■■ SYNTHETIC DATA NOTICE

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This project uses a synthetic student attendance dataset created for academic demonstration. It does not contain real student names, roll numbers, email IDs, or actual attendance records of identifiable students. All records are computer-generated using statistical modeling. No real student information was collected, stored, or processed at any stage.

Terminology Notice

Early draft documentation (PROJECT_PLAN.md) used the terms "**Safe**" and "**At Risk**". All data files, scripts, validation reports, model outputs, and this report use the following consistent terminology:

Term	Equivalent in Early Drafts	Definition
Regular	Safe	Attendance_Percentage $\geq$ 75%
Defaulter	At Risk	Attendance_Percentage $<$ 75%

This mapping is documented once here and used consistently throughout this report.

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1. Abstract

This project presents a complete end-to-end data science pipeline for student attendance analysis and prediction using a fully synthetic, privacy-preserving dataset. The dataset comprises 4,100 records representing 205 anonymous students (STU0001–STU0205) across two departments — Computer Engineering and MCA — over five subjects and four attendance periods per subject. All student identifiers are synthetic; no real student data is involved at any stage.

The project implements two machine learning tasks: (1) **Regression** — predicting a student's continuous attendance percentage, and (2) **Classification** — predicting whether a student is **Regular** (Attendance_Percentage $\geq$ 75%) or a **Defaulter** (Attendance_Percentage $<$ 75%).

After systematic model comparison, **GradientBoostingRegressor** achieved the best regression performance (RMSE: 11.3952, R^2 : 0.6874) and **GradientBoostingClassifier** achieved the best classification performance (F1-Score: 0.8606, ROC-AUC: 0.9189, Recall: 0.8587). The project demonstrates responsible, ethical data science practices while delivering a working predictive system suitable for academic portfolio and viva demonstration.

2. Introduction

Student attendance is widely recognized as a critical indicator of academic engagement and performance outcomes. Educational institutions maintain attendance registers to enforce minimum attendance requirements — typically set at 75% — below which students may be barred from examinations or required to seek remedial intervention.

However, building data-driven attendance prediction models using actual student records raises significant privacy and ethical concerns. Real attendance datasets contain personally identifiable information (PII) such as student names, roll numbers, and institutional email IDs. Handling such data requires explicit institutional consent, data governance compliance, and careful anonymization — constraints that are difficult to fulfill in a student project setting.

This project resolves that tension by working with a **fully synthetic dataset**. The synthetic data was statistically modeled to mirror the realistic distributions and correlations of a genuine classroom attendance dataset, without capturing any real student information. Using this dataset, the project demonstrates that a complete, reproducible, and publication-quality data science workflow can be executed ethically and responsibly.

The project covers data generation, validation, exploratory analysis, regression modeling, classification modeling, and a final interpretive summary — making it a complete academic data science project

demonstrating modern ML practices.

3. Problem Statement

Student absenteeism is a persistent challenge in higher education. Students who fall below the minimum required attendance threshold risk academic penalties, poor exam performance, and increased dropout rates. Faculty and administration often identify attendance problems only after significant deterioration has occurred, leaving limited time for corrective intervention.

The core problem is twofold:

1. **Prediction Problem (Regression):** Given a student's demographic profile, prior

academic performance, and study habits, can we predict their likely attendance percentage for the current semester?

2. **Classification Problem (Early Warning):** Can we classify a student as **Regular**

(Attendance $\geq$ 75%) or **Defaulter** (Attendance $<$ 75%) early enough to trigger meaningful academic intervention?

A secondary challenge is doing this responsibly — without requiring access to real student PII — by demonstrating that synthetic data can serve as a viable and ethically sound substitute for model development and academic exploration.

4. Objectives

#	Objective
1	Generate a fully synthetic, anonymous student attendance dataset that preserves realistic statistical properties
2	Validate the synthetic dataset against 25 data quality, integrity, and privacy checks

#	Objective
3	Perform comprehensive Exploratory Data Analysis (EDA) to uncover attendance patterns and feature correlations
4	Build and evaluate multiple Regression models to predict Attendance_Percentage (continuous)
5	Build and evaluate multiple Classification models to predict Attendance_Status (Regular / Defaulter)
6	Select the best model for each task using principled metric-based criteria
7	Save trained model pipelines for reusable deployment
8	Demonstrate responsible, ethical, and privacy-preserving data science practices throughout

5. Scope

In Scope:

- Synthetic data generation, validation, and documentation
- Exploratory Data Analysis with 11 professional visualizations
- Regression modeling (4 algorithms, full pipeline with preprocessing)
- Classification modeling (4 algorithms, full pipeline with preprocessing)
- Model evaluation, comparison, and best-model selection
- Saved, deployable model pipelines
- Complete summary report and presentation

Out of Scope:

- Real student data of any kind
- Real-time deployment or integration with institutional LMS/ERP systems
- FERPA/institutional data governance compliance (not applicable to synthetic data)
- Causal inference or interventional studies
- Time-series attendance modeling (week-by-week trends)

- Multi-institution comparison

6. Synthetic Dataset Description

6.1 Generation Overview

The dataset was generated in two phases using NumPy random seeding for reproducibility:

- **Phase 1:** 60 students (STU0001–STU0060), Computer Engineering, Third Year,

Fifth Semester — generated by `src/generate_dataset.py` (seed=42)

- **Phase 2 Extension:** 145 additional students (STU0061–STU0205), MCA, Final Year,

Third Semester — added by `src/extend_dataset_to_205.py` (seed=42+200)

6.2 Final Dataset Parameters

Parameter	Value
File	<code>data/student_attendance_205_students.csv</code>
Total Rows	4,100
Total Columns	21
Unique Students	205 (STU0001–STU0205)
Rows per Student	20 (5 subjects × 4 attendance periods)
Unique Subjects	5
Departments	Computer Engineering (1,200 records), MCA (2,900 records)
Attendance Range	0.00% – 100.00%
Mean Attendance	68.67%
Median Attendance	70.00%
Std Deviation	20.21%

Parameter	Value
Regular Records	1,834 (44.73%)
Defaulter Records	2,266 (55.27%)
Missing Values	0
Duplicate Rows	0

6.3 Generation Strategy

- Attendance values generated using a clipped normal distribution (mean ~75%, std ~15%)
- Subject-level mild correlation applied so students who attend one subject tend to attend others
- All values rounded and clipped to valid integer range [0, Total_Classes]
- Attendance_Percentage computed as $(\text{Classes_Attended} / \text{Total_Classes}) \times 100$
- Attendance_Status assigned by rule: $\geq 75\% \rightarrow \text{Regular}$; $< 75\% \rightarrow \text{Defaulter}$
- Student IDs use the format STU0001–STU0205 — no real identifiers

6.4 Subjects

Code	Subject Name
1	Computer Networks
2	Data Structures & Algorithms
3	Database Management Systems
4	Software Engineering
5	Theory of Computation

6.5 Validation

The final dataset was validated by `src/validate_final_dataset.py`, which executed **25 independent checks** covering file existence, row/column counts, student ID integrity, subject completeness, missing values, duplicates, attendance calculation accuracy, status label correctness, numeric range validity, and absence of private data patterns.

Result: 25/25 checks PASSED.

Full report: `outputs/final_dataset_validation_report.txt`

7. Privacy and Ethical Approach

Principle	Implementation
Data Minimization	Only statistical classroom parameters used; no personal data collected or stored
Full Anonymization	All students assigned synthetic IDs (STU0001–STU0205); no real names or roll numbers exist in any file
No Re-identification Risk	No mapping table between synthetic IDs and real students was ever created
Synthetic Data Labeling	Every output file, notebook, and report clearly labels data as SYNTHETIC
Honest Framing	All conclusions explicitly acknowledge the synthetic nature of the data
Source Isolation	The <code>private_original_data/</code> directory is physically present but never read, imported, or inspected by any project script
No Discrimination	The Defaulter label is used solely for academic analysis — never for real-world academic judgment
Reproducibility	Fixed random seeds (42) ensure anyone can reproduce the exact same dataset

Important for Real-World Deployment: Before applying any attendance prediction model to real student data, institutions must obtain informed consent, comply with applicable education data privacy regulations, and ensure human oversight before acting on model-generated flags.

8. Dataset Columns

#	Column	Data Type	Description
1	<code>Student_ID</code>	Categorical	Anonymous synthetic ID (STU0001–STU0205)

#	Column	Data Type	Description
2	Gender	Categorical	Male / Female
3	Age	Numeric	Student age (integer)
4	Department	Categorical	Computer Engineering / MCA
5	Year	Categorical	Third Year / Final Year
6	Semester	Categorical	Fifth Semester / Third Semester
7	Subject	Categorical	One of the 5 subjects listed in Section 6.4
8	Attendance_Period	Categorical	Period_1 / Period_2 / Period_3 / Period_4
9	Total_Classes	Numeric	Total classes scheduled in that period (8–12)
10	Classes_Attended	Numeric	Classes the student actually attended (0–12)
11	Previous_Attendance_Percentage	Numeric	Student's prior semester attendance %
12	Assignment_Score	Numeric	Score on course assignments
13	Internal_Marks	Numeric	Internal/mid-term exam marks
14	Study_Hours_Per_Week	Numeric	Self-reported weekly study hours
15	Medical_Leave_Days	Numeric	Days of medical leave taken this semester
16	Travel_Distance_KM	Numeric	Distance between student's home and college
17	Previous_Exam_Score	Numeric	Score in the previous semester's final exam

#	Column	Data Type	Description
18	Late_Count	Numeric	Number of late arrivals to class
19	Online_Class_Attendance	Numeric	Online/virtual session attendance percentage
20	Attendance_Percentage	Numeric	Regression Target: (Classes_Attended / Total_Classes) × 100
21	Attendance_Status	Categorical	Classification Target: Regular (≥75%) / Defaulter (<75%)

Note: **Classes_Attended** and **Total_Classes** (columns 9–10) are mathematically related to **Attendance_Percentage** (column 20). These two columns were **excluded from all model features** to prevent data leakage.

9. Data Preprocessing

Both the regression and classification pipelines use an identical preprocessing workflow implemented as a **sklearn.pipeline.Pipeline** combined with a **ColumnTransformer**.

9.1 Feature Selection

15 predictors used (no leakage features):

Categorical (5): Gender, Department, Year, Semester, Subject

Numeric (10): Age, Previous_Attendance_Percentage, Assignment_Score, Internal_Marks, Study_Hours_Per_Week, Medical_Leave_Days, Travel_Distance_KM, Previous_Exam_Score, Late_Count, Online_Class_Attendance

Excluded (data leakage): Classes_Attended, Total_Classes, Attendance_Period, Student_ID

9.2 Preprocessing Steps

Step	Categorical Features	Numeric Features
Imputation	<code>SimpleImputer(strategy='most_frequent')</code>	<code>SimpleImputer(strategy='median')</code>
Encoding / Scaling	<code>OneHotEncoder(handle_unknown='ignore')</code>	<code>StandardScaler()</code>

9.3 Train-Test Split

Parameter	Value
Split ratio	80% train / 20% test
Random state	42
Stratification	Applied for classification (preserves class ratios)
Training rows	~3,280
Test rows	~820

10. Exploratory Data Analysis

EDA was performed in `notebooks/01_exploratory_data_analysis.ipynb` using `src/exploratory_analysis.py`. Nine visualizations were generated and saved to `outputs/charts/`.

10.1 Attendance Distribution

The overall attendance percentage follows a near-normal distribution with:

- Mean: **68.67%**, Median: **70.00%**, Std Dev: **20.21%**
- Distribution peaks between 70–80%
- A secondary concentration of zero-attendance records exists (students who missed entire periods)

- The 75% threshold divides the dataset into Regular (44.73%) and Defaulter (55.27%)

10.2 Regular vs. Defaulter Distribution

Status	Count	Percentage
Defaulter (Attendance < 75%)	2,266	55.27%
Regular (Attendance ≥ 75%)	1,834	44.73%

The majority class is Defaulter. This slight imbalance was addressed in classification by using stratified train-test split and prioritizing F1-score and Recall over raw accuracy.

10.3 Department-wise Analysis

Department	Records	Mean Attendance	Median Attendance
Computer Engineering	1,200	70.89%	72.73%
MCA	2,900	67.75%	66.67%

Computer Engineering students show slightly higher mean attendance. This is consistent with the dataset generation parameters which used different base rates for the two cohorts.

10.4 Subject-wise Analysis

Subject	Records	Mean Attendance	Median Attendance
Software Engineering	820	69.12%	70.00%
Computer Networks	820	68.86%	71.37%
Data Structures & Algorithms	820	68.54%	70.00%
Theory of Computation	820	68.43%	70.00%
Database Management Systems	820	68.38%	66.67%

Subject-level variation is minimal (< 1%). Attendance behaviour is predominantly student-level, not subject-driven.

10.5 Feature Correlation Analysis (Pearson)

Feature	Correlation with Attendance_Percentage	Interpretation
Classes_Attended	0.90	Mathematical relationship (excluded)
Previous_Attendance_Percentage	0.72	Strong Positive
Internal_Marks	0.67	Strong Positive
Study_Hours_Per_Week	0.61	Moderate-Strong Positive
Previous_Exam_Score	0.59	Moderate-Strong Positive
Assignment_Score	0.37	Moderate Positive
Online_Class_Attendance	-0.00	No Correlation
Travel_Distance_KM	0.04	No Correlation
Late_Count	0.01	No Correlation
Medical_Leave_Days	-0.02	No Correlation
Age	-0.07	Negligible Negative

Key Insight: Prior academic behavior (**Previous_Attendance_Percentage**, **Internal_Marks**, **Previous_Exam_Score**) is the dominant predictor group. Logistical and personal factors (travel distance, medical leave, late count) show no meaningful correlation in this synthetic dataset.

10.6 Correlation vs. Causation

Statistical correlations observed in this dataset **do not imply causal relationships**. A positive correlation of 0.67 between **Internal_Marks** and **Attendance_Percentage** does not mean that higher attendance causes better marks — both could be driven by underlying student motivation, diligence, or academic support systems. This is especially important to note in a synthetic dataset where correlations are mathematically

encoded during generation, not observed from natural behavior.

10.7 Visualizations Generated

File	Description
attendance_distribution.png	KDE histogram with 75% threshold line
regular_defaulter_count.png	Bar chart: Regular vs Defaulter counts
subject_wise_attendance.png	Boxplot per subject
attendance_by_period.png	Boxplot across Period_1 to Period_4
study_hours_vs_attendance.png	Scatter: Study hours vs attendance
internal_marks_vs_attendance.png	Scatter: Internal marks vs attendance
medical_leave_vs_attendance.png	Boxplot: Medical leave vs attendance
late_count_vs_attendance.png	Scatter: Late count vs attendance
correlation_heatmap.png	Pearson correlation matrix heatmap
regression_actual_vs_predicted.png	Actual vs predicted (best regressor)
best_classifier_confusion_matrix.png	Confusion matrix (best classifier)

11. Regression Methodology

11.1 Task Definition

- **Target variable:** `Attendance_Percentage` (continuous, range: 0–100)
- **Type:** Supervised Regression
- **Script:** `src/train_regression.py`
- **Notebook:** `notebooks/02_regression_model.ipynb`

11.2 Pipeline Architecture

Raw Data → ColumnTransformer ■■■ Numeric Branch: `SimpleImputer(median)` → `StandardScaler` ■■■ Categorical Branch: `SimpleImputer(most_frequent)` → `OneHotEncoder` → Regressor Model → Predicted `Attendance_Percentage`

11.3 Models Trained

Model	Key Hyperparameters
LinearRegression	Default
DecisionTreeRegressor	<code>max_depth=8</code> , <code>random_state=42</code>
RandomForestRegressor	<code>n_estimators=100</code> , <code>max_depth=12</code> , <code>random_state=42</code>
GradientBoostingRegressor	<code>n_estimators=100</code> , <code>learning_rate=0.1</code> , <code>random_state=42</code>

12. Classification Methodology

12.1 Task Definition

- **Target variable:** `Attendance_Status` (Regular=0, Defaulter=1)
- **Type:** Supervised Binary Classification
- **Script:** `src/train_classification.py`
- **Notebook:** `notebooks/03_classification_model.ipynb`

12.2 Target Encoding

Label	Encoded Value
Regular	0
Defaulter	1

12.3 Pipeline Architecture

Raw Data → ColumnTransformer (same as Regression) → Classifier Model → Predicted Attendance_Status (Regular / Defaulter) + Predicted Probabilities (for ROC-AUC)

12.4 Models Trained

Model	Key Hyperparameters
LogisticRegression	max_iter=1000, random_state=42
DecisionTreeClassifier	max_depth=6, random_state=42
RandomForestClassifier	n_estimators=100, max_depth=10, random_state=42
GradientBoostingClassifier	n_estimators=100, learning_rate=0.1, random_state=42

13. Algorithms Used

13.1 Gradient Boosting (Both Best Models)

Gradient Boosting is an ensemble learning method that builds models sequentially. Each new tree corrects the residual errors of all preceding trees:

$$F_m(x) = F_{\{m-1\}}(x) + \eta \cdot h_m(x)$$

Where $F_m(x)$ is the model at iteration m , η is the learning rate (0.1), and $h_m(x)$ is a decision tree fitted on the residuals of $F_{\{m-1\}}(x)$.

Why it performs well here:

- Handles mixed categorical + numeric features effectively

- Captures non-linear relationships between attendance and predictors
- Sequential error correction reduces both bias and variance
- Less prone to overfitting than single deep trees

13.2 Random Forest

An ensemble of independently trained decision trees that vote (classification) or average (regression) their predictions. Uses bootstrap sampling and random feature selection at each split to reduce variance.

13.3 Decision Tree

A single recursive binary splitting tree. Interpretable but prone to overfitting on complex datasets, hence typically outperformed by ensembles.

13.4 Linear / Logistic Regression

Linear Regression fits a hyperplane minimizing mean squared error. Logistic Regression models class probability using the sigmoid function. Both serve as interpretable baselines.

14. Evaluation Metrics

14.1 Regression Metrics

Metric	Formula	Interpretation		
MAE	$\text{mean}(y - \hat{y})$	Average absolute error		Average absolute error in attendance percentage points
MSE	$\text{mean}((y - \hat{y})^2)$	Mean squared error (penalizes large errors more)		
RMSE	$\sqrt{\text{MSE}}$	Root MSE — same units as target (percentage points)		

Metric	Formula	Interpretation		
R^2	$1 - SS_{res}/SS_{tot}$	Proportion of variance explained (1.0 = perfect)		

Primary selection criterion: Lowest RMSE + Highest R^2

14.2 Classification Metrics

Metric	Formula	Interpretation
Accuracy	$(TP+TN)/(TP+TN+FP+FN)$	Overall correct predictions
Precision	$TP/(TP+FP)$	Of predicted Defaulters, how many are actually Defaulters
Recall	$TP/(TP+FN)$	Of actual Defaulters, how many were correctly identified
F1-Score	$2 \cdot \text{Prec} \cdot \text{Rec} / (\text{Prec} + \text{Rec})$	Harmonic mean of Precision and Recall
ROC-AUC	Area under ROC curve	Ability to discriminate Regular from Defaulter across all thresholds

Primary selection criterion: Highest F1-Score, then highest Recall

Recall is prioritized in the secondary criterion because in an early-warning system, a **False Negative** (failing to identify a Defaulter) is more costly than a **False Positive** (incorrectly flagging a Regular student).

15. Actual Regression Results

Source: `outputs/regression_model_results.csv` — values verified directly from file

Model	MAE	MSE	RMSE	R^2
LinearRegression	9.3679	134.8109	11.6108	0.6755

Model	MAE	MSE	RMSE	R ²
DecisionTreeRegressor	10.2138	164.6423	12.8313	0.6037
RandomForestRegressor	9.1673	130.7760	11.4357	0.6852
GradientBoostingRegressor	9.1786	129.8510	11.3952	0.6874

Full precision values (from CSV, unrounded):

Model	MAE (full)	MSE (full)	RMSE (full)	R ² (full)
LinearRegression	9.367864124465022	134.8109039111988	11.610809787056146	0.6754938609166184
DecisionTreeRegressor	10.213813496653357	164.642279552681	12.831300773993298	0.6036861343002312
RandomForestRegressor	9.167304377304418	130.77602261653715	11.435734458990256	0.6852063078523121
GradientBoostingRegressor	9.178560849121798	129.8509839512371	11.395217591219446	0.6874329877207083

Chart: [outputs/charts/regression_actual_vs_predicted.png](#)

16. Actual Classification Results

Source: [outputs/classification_model_results.csv](#) — values verified directly from file

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
LogisticRegression	0.8390	0.8512	0.8587	0.8549	0.9246
DecisionTreeClassifier	0.8268	0.8659	0.8124	0.8383	0.8841

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
RandomForest Classifier	0.8451	0.8655	0.8521	0.8587	0.9276
GradientBoostingClassifier	0.8463	0.8625	0.8587	0.8606	0.9189

Full precision values (from CSV, unrounded):

Model	Accuracy (full)	Precision (full)	Recall (full)	F1-Score (full)	ROC-AUC (full)
LogisticRegression	0.8390243902439024	0.8512035010940919	0.8587196467991169	0.8549450549450549	0.9245778972758059
DecisionTreeClassifier	0.8268292682926829	0.8658823529411764	0.8123620309050773	0.8382687927107062	0.8840819002592465
RandomForest Classifier	0.8451219512195122	0.8654708520179372	0.8520971302428256	0.8587319243604005	0.9276214879910496
GradientBoostingClassifier	0.8463414634146341	0.8625277161862528	0.8587196467991169	0.8606194690265486	0.9188937209400244

Chart: [outputs/charts/best_classifier_confusion_matrix.png](#)

17. Best Model Selection

17.1 Best Regression Model — GradientBoostingRegressor

Selection Criterion: Lowest RMSE (and correspondingly highest R²)

Model	RMSE	R ²	Rank
GradientBoostingRegressor	11.3952	0.6874	1 ■
RandomForestRegressor	11.4357	0.6852	2
LinearRegression	11.6108	0.6755	3

Model	RMSE	R ²	Rank
DecisionTreeRegressor	12.8313	0.6037	4

Rationale:

- Achieved lowest RMSE (11.3952) and highest R² (0.6874) simultaneously
- GradientBoosting's sequential error-correction particularly benefits mixed feature datasets
- Margin over RandomForest is small but consistent (RMSE 0.04 lower, R² 0.0022 higher)
- DecisionTree overfits — highest error; LinearRegression assumes linearity that doesn't fully hold

Model saved: `models/best_regression_model.joblib`

Interpretation: R² = 0.6874 means the model explains **68.74%** of variance in student attendance percentage. The remaining unexplained variance reflects features with near-zero correlation (travel distance, medical leave, late count) that were included for completeness but contribute minimal predictive signal.

17.2 Best Classification Model — GradientBoostingClassifier

Selection Criterion: Highest F1-Score, then Recall (accuracy alone is insufficient due to class imbalance and early-warning context)

Model	F1-Score	Recall	Accuracy	ROC-AUC	Rank
GradientBoostingClassifier	0.8606	0.8587	0.8463	0.9189	1 ■
RandomForestClassifier	0.8587	0.8521	0.8451	0.9276	2
LogisticRegression	0.8549	0.8587	0.8390	0.9246	3
DecisionTreeClassifier	0.8383	0.8124	0.8268	0.8841	4

Rationale:

- Highest F1-Score (0.8606) balances Precision and Recall optimally
- Equal Recall to LogisticRegression (0.8587) but higher F1 and Accuracy

- RandomForest has higher ROC-AUC (0.9276) but lower F1-Score — ROC-AUC is not the primary criterion for this early-warning use case
- DecisionTree shows worst Recall (0.8124) — unacceptable for an early-warning system

Model saved: `models/best_classification_model.joblib`

Interpretation:

- **Precision (86.25%):** When the model flags a student as Defaulter, it is correct 86.25% of the time
- **Recall (85.87%):** The model correctly identifies 85.87% of all actual Defaulters
- **F1-Score (0.8606):** Strong balance between identifying Defaulters and avoiding false alarms
- **ROC-AUC (0.9189):** Excellent discriminating power across all classification thresholds

18. Sample Prediction

The following demonstration uses the saved model pipelines to predict for a hypothetical student. This student is not part of the training or test data.

Sample Student Profile:

Feature	Value
Gender	Male
Department	Computer Engineering
Year	Third Year
Semester	Fifth Semester
Subject	Data Structures & Algorithms
Age	20
Previous_Attendance_Percent age	68.5%

Feature	Value
Assignment_Score	62.0
Internal_Marks	15.0
Study_Hours_Per_Week	10.0
Medical_Leave_Days	2
Travel_Distance_KM	8.5
Previous_Exam_Score	58.0
Late_Count	1
Online_Class_Attendance	70.0%

Actual prediction output can be reproduced by running the code cell in [notebooks/05_summary_report.ipynb](#) Section 11.

```
Loading the models: `python import joblib best_regressor =  
joblib.load("models/best_regression_model.joblib") best_classifier =  
joblib.load("models/best_classification_model.joblib")`
```

Predict

```
predicted_pct = best_regressor.predict(sample_student)[0] predicted_status = "Defaulter" if  
best_classifier.predict(sample_student)[0] == 1 else "Regular" predicted_proba =  
best_classifier.predict_proba(sample_student)[0] `
```

19. Findings

19.1 Dataset Findings

1. The 205-student dataset (4,100 records) is fully clean — 0 missing values, 0 duplicates,

25/25 validation checks passed 2. **55.27%** of records fall below the 75% threshold (Defaulter) — the threshold is genuinely challenging 3. Mean attendance (**68.67%**) is below the 75% cutoff, indicating that the average student in this dataset is a Defaulter

19.2 EDA Findings

4. Subject-level attendance variation is minimal (< 1%) — attendance behaviour is primarily

student-level, not subject-driven 5. Computer Engineering students show higher mean attendance (70.89%) than MCA (67.75%) 6. Prior academic behavior dominates as a predictor:

Previous_Attendance_Percentage (r=0.72) and **Internal_Marks** (r=0.67) are the two strongest correlates 7. Logistical factors (**Travel_Distance_KM**, **Late_Count**, **Medical_Leave_Days**) show no meaningful correlation — these do not systematically drive attendance in this dataset

19.3 Modelling Findings

8. **GradientBoostingRegressor** is the best regressor with RMSE=11.3952 and R²=0.6874

9. **GradientBoostingClassifier** is the best classifier with F1=0.8606, Recall=0.8587

10. Linear models (LinearRegression, LogisticRegression) are competitive — they are

interpretable baselines that perform near the level of more complex ensembles 11. DecisionTree models (both regressor and classifier) show the weakest performance, suggesting single-tree models overfit on this dataset

20. Practical Applications

If deployed on real (properly anonymized, consent-compliant) student data:

Use Case	Description
Early Semester Warning	Predict attendance trajectory after the first few weeks; trigger counseling before the attendance falls irreversibly below 75%
Department-Level Analytics	HOD can view department-wide attendance risk distribution; allocate faculty support accordingly
Class Teacher Dashboard	Teachers receive automated flags for students predicted as Defaulters in their specific subject
Student Self-Assessment	Students can input their own profile and receive a predicted attendance status — proactive self-monitoring
Institutional Policy Design	Analyze which feature groups (academic engagement vs. logistical factors) are most predictive; inform policy

Deployment Pipeline: Attendance Data → Preprocessing → Trained Pipeline → Prediction
→ Alert / Report

21. Limitations

#	Limitation	Impact Level
1	Synthetic Data — Results are statistically modeled, not empirically observed from real student behavior	High
2	No Causal Analysis — Correlations are identified, not causal mechanisms	Medium
3	No Temporal Component — Attendance is recorded as period-level totals, not as a week-by-week time series	Medium
4	No External Factors — Health status, family income, mental health, transportation access are not modeled	Medium

#	Limitation	Impact Level
5	Fixed Threshold — The 75% threshold is used as a universal rule; real institutions may apply different criteria	Low
6	Model Generalizability — A model trained on synthetic data cannot be directly deployed on real student data without retraining	High
7	Dataset Size — 205 students is a moderate size; larger datasets would improve model stability	Low
8	Leakage Exclusion Trade-off — Excluding Classe s_Attended and Total_Classes is necessary but reduces R^2	Medium

22. Future Scope

Direction	Description
Time-Series Modeling	Use LSTM, Transformer, or Prophet on week-by-week attendance sequences to detect deterioration early in the semester

Direction	Description
Real Data Integration	Retrain the complete pipeline on real, properly anonymized, consent-based student attendance data for institutional deployment
Explainability (SHAP)	Apply SHAP values to explain individual predictions — why a specific student was flagged as Defaulter
Multi-Class Classification	Expand to 3-class model: Regular / At-Risk / Critical-Defaulter for more granular early-warning
Web Dashboard	Build a Streamlit or Flask-based dashboard for real-time prediction and visualization
Differential Privacy	Apply differential privacy mechanisms to formal mathematical guarantees during synthetic generation
Cross-Cohort Benchmarking	Extend the dataset to additional departments, years, and institution types
Active Learning	Identify the most uncertain predictions and design targeted data collection to improve model accuracy

23. Conclusion

This project demonstrates a complete, reproducible, and ethically responsible data science workflow for student attendance analysis and prediction. Working entirely with a synthetic dataset eliminates privacy risks while preserving the ability to explore realistic attendance patterns and build functional predictive models.

Key accomplishments:

- Generated and validated a clean, realistic 205-student synthetic attendance dataset

(4,100 records, 25/25 validation checks passed)

- Conducted thorough EDA revealing that prior academic behavior is the dominant predictor

of attendance, with logistical factors showing negligible correlation

- Trained and evaluated 8 machine learning models (4 regression + 4 classification)

using rigorous preprocessing pipelines and principled evaluation criteria

- Selected **GradientBoostingRegressor** as the best regression model (RMSE: 11.3952,

R^2 : 0.6874) and **GradientBoostingClassifier** as the best classification model (F1-Score: 0.8606, ROC-AUC: 0.9189)

- Saved deployable model pipelines and created comprehensive documentation

The project establishes that **Gradient Boosting ensemble methods** consistently outperform simpler models on mixed-feature attendance datasets. The classification model's Recall of **85.87%** makes it suitable for an early-warning system where correctly identifying Defaulters is the highest priority.

Privacy Assurance: This project used zero real student data at any stage. All records are computer-generated. The `private_original_data/` directory was isolated and never accessed. This project is safe for academic submission, public portfolio sharing, and viva demonstration.

24. References

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Annals of Statistics, 29(5), 1189–1232. 3. Pedregosa, F. et al. (2011). *Scikit-learn: Machine Learning in Python*. Journal of Machine Learning Research, 12, 2825–2830. 4. McKinney, W. (2010). *Data Structures for Statistical Computing in Python*. Proceedings of the 9th Python in Science Conference, 51–56. 5. Waskom, M. (2021). *Seaborn: Statistical Data Visualization*. Journal of Open Source Software, 6(60), 3021. 6. Harris, C. R. et al. (2020). *Array Programming with NumPy*. Nature, 585, 357–362. 7. European Commission. (2018). *General Data Protection Regulation (GDPR)*. Official Journal of the European Union, L 119/1. 8. Dwork, C. (2006). *Differential Privacy*. International Colloquium on Automata, Languages and Programming, 1–12. 9. Tinto, V. (1975). *Dropout from Higher Education: A Theoretical Synthesis of Recent Research*. Review of Educational Research, 45(1), 89–125. 10. scikit-learn Documentation. (2024). *Pipeline, ColumnTransformer, GradientBoosting*. Retrieved from <https://scikit-learn.org/stable/>

25. Appendix

A. File Structure

Data_Science_attendence_project/ ■■■ data/ ■ ■■■
student_attendance_205_students.csv [Main dataset – 4,100 rows × 21 cols] ■ ■■■
student_attendance_205_students.xlsx ■ ■■■ student_attendance.csv [Original
60-student dataset] ■ ■■■ DATASET_DICTIONARY.md ■ ■■■ SYNTHETIC_DATA_NOTICE.md
■■■ models/ ■ ■■■ best_regression_model.joblib [GradientBoostingRegressor
pipeline] ■ ■■■ best_classification_model.joblib [GradientBoostingClassifier
pipeline] ■■■ notebooks/ ■ ■■■ 01_exploratory_data_analysis.ipynb ■ ■■■
02_regression_model.ipynb ■ ■■■ 03_classification_model.ipynb ■ ■■■
05_summary_report.ipynb ■■■ outputs/ ■ ■■■ charts/ [11 PNG visualization files]
■ ■■■ regression_model_results.csv ■ ■■■ classification_model_results.csv ■ ■■■
final_model_comparison.csv ■ ■■■ final_project_summary.md ■ ■■■
final_findings.txt ■ ■■■ eda_summary.md ■ ■■■
final_dataset_validation_report.txt ■ ■■■ dataset_validation_report.txt ■■■
report/ ■ ■■■ project_report.md [This file] ■ ■■■ project_report.pdf [PDF
version – see conversion note] ■■■ presentation/ ■ ■■■ presentation_content.md ■
■■■ project_presentation.pptx ■ ■■■ viva_questions.md ■■■ src/ ■ ■■■
generate_dataset.py ■ ■■■ extend_dataset_to_205.py ■ ■■■ validate_dataset.py ■
■■■ validate_final_dataset.py ■ ■■■ exploratory_analysis.py ■ ■■■
train_regression.py ■ ■■■ train_classification.py ■ ■■■ generate_eda_notebook.py
■ ■■■ generate_regression_notebook.py ■ ■■■ generate_classification_notebook.py
■ ■■■ generate_summary_notebook.py ■■■ private_original_data/ [ISOLATED – never
accessed by scripts] ■■■ PROJECT_PLAN.md ■■■ PROJECT_STATUS.md

B. Random Seeds Used

Script	Seed	Purpose
generate_dataset.py	42	60-student base dataset
extend_dataset_to_205.py	42 + 200 = 242	MCA extension dataset
All model training	42	Train-test split and model initialization

C. Python Environment

Package	Version (from PROJECT_STATUS.md)
Python	3.13.7
pandas	2.3.3
numpy	2.3.3

Package	Version (from PROJECT_STATUS.md)
scikit-learn	Installed (version per pip freeze)
matplotlib	Installed
seaborn	Installed
joblib	Installed
openpyxl	3.1.5

D. Metrics Verification

All metric values in this report were verified directly against their source CSV files:

- Regression metrics: **outputs/regression_model_results.csv**
- Classification metrics: **outputs/classification_model_results.csv**

No values were estimated, rounded incorrectly, or fabricated. Full-precision values are provided in Sections 15 and 16.